

Text Representation

Learning Objectives

- Understand ASCII encoding for English text
- Explain how Unicode supports multiple languages
- Convert text to binary using character codes

Engage (5-7 minutes)

Think of a code language you use with friends—each letter 替换为 a number only you understand. Computers do the same with text—each letter becomes a number. Today we decode how computers read words.

Explore (10-12 minutes)

Look up ASCII values for your name—A=65, B=66, a=97, b=98. Write the binary for each letter in your name. Count how many bits are needed for one character. Discuss why computers need codes for letters.

Explain (10-12 minutes)

ASCII maps 128 characters to numbers—uppercase A-Z (65-90), lowercase a-z (97-122), digits 0-9 (48-57). Unicode extends this to include Hindi, Kannada, and every world script. Each character has a unique code point that computers store as binary.

Elaborate (8-10 minutes)

Write your name in ASCII decimal values, then convert each to binary. For example, “RAVI” becomes 82 65 86 73 in decimal. Discuss why English text needs 7-8 bits per character while Kannada might need more.

Evaluate (5-8 minutes)

Exit ticket: What does ASCII stand for? Write the ASCII decimal value for letter ‘K’ and its binary equivalent.